



CS458 GAME DEVELOPMENT PROJECT

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# Flappy Bean Game Design Document

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# Game Title

Flappy Bean

## Game Idea

Flappy Bean is a side-scrolling endless runner game where the player controls a coffee bean navigating through obstacles. The game is designed to provide a quick, challenging, and addictive experience similar to the classic “Flappy Bird” but with a unique coffee-themed twist. The selection of this game is motivated by the desire to create a simple yet engaging game that appeals to casual gamers and coffee lovers alike.

## Target Audience

The target audience for Flappy Bean includes:

- **Casual Gamers:** Players who enjoy quick sessions of gameplay during breaks or commute.
- **Coffee Enthusiasts:** People who appreciate coffee culture and would find the theme relatable and amusing.
- **Competitive Players:** Gamers who enjoy high-score chasing and challenging their friends.

This audience was selected because the game’s mechanics are easy to pick up but hard to master, making it accessible to a wide range of players while offering depth for those who want to compete.

## Game Genre

**Genre:** Endless Runner / Arcade

Flappy Bean falls under the Endless Runner and Arcade genres because:

- **Endless Gameplay:** The game continues indefinitely until the player fails.
- **Score-Based:** The primary goal is to achieve the highest possible score.
- **Simple Controls:** The game uses a single input (tap/click) to control the character.
- **Increasing Difficulty:** The game becomes progressively more challenging as the player advances.

# Character

**Character:** A sentient Coffee Bean.

**Motivation:** The character is a coffee bean who suddenly gained consciousness. The motivation for selecting this character is to create a humorous and relatable protagonist who is trying to escape the fate of being ground and consumed. This adds a layer of narrative and personality to the game.

## Gameplay

The gameplay revolves around navigating the coffee bean through a series of obstacles (pipes) without colliding with them or the ground.

## Challenges

- **Obstacles:** The primary challenge is to fly between pipes that are placed at varying heights.
- **Gravity:** The player must constantly tap to counteract gravity and keep the bean afloat.
- **Precision:** The gap between pipes requires precise timing and control to pass through safely.

## Actions

- **Jump/Flap:** The player can tap the screen or press a key to make the bean flap its “wings” and gain altitude.
- **Fall:** When not flapping, the bean falls due to gravity.

## Challenge Type

- **Skill-Based:** The game relies heavily on the player’s reaction time and hand-eye coordination.
- **Endurance:** The challenge is to maintain focus and consistency over time.

## Termination Condition

The game ends when:

- The bean collides with a pipe.
- The bean hits the ground.
- The time limit expires.

## Rewards

Rewards are provided to the player to encourage progression and enhance the gameplay experience.

- **Coins:** Players collect coins scattered throughout the level. These can be used to unlock new skins.
- **Medals/Cups:**
  - **Silver Cup:** Awarded for passing a certain number of pipes (e.g., 10).
  - **Gold Cup:** Awarded for passing a higher number of pipes (e.g., 20).
- **Shield:** A power-up that allows the bird to survive crashes for a limited time after collecting a specified number of coins.

## Future Work

To further improve the game, the following features could be added:

- **Suction Pipes:** Introduce pipes that create a vacuum effect, trying to pull the player into them, adding a new layer of difficulty.
- **New Characters:** Add more coffee-themed characters (e.g., Espresso Shot, Cappuccino Foam).
- **Power-ups:** Introduce more power-ups like speed boosts or size reduction.
- **Leaderboards:** Implement a global leaderboard to foster competition among players.